

Client server Basics

(This room teaches Basics of client server model):-

Task 1:- Introduction

Most computers work alone; they stored their own files, ran their own programs, and did not ^{communicate with other computers}.
Soon, multiple organization around world started with idea of interconnecting these systems, to facilitate information exchange and resource sharing. Hence, the predecessor of Internet was born.

Learning objectives :-

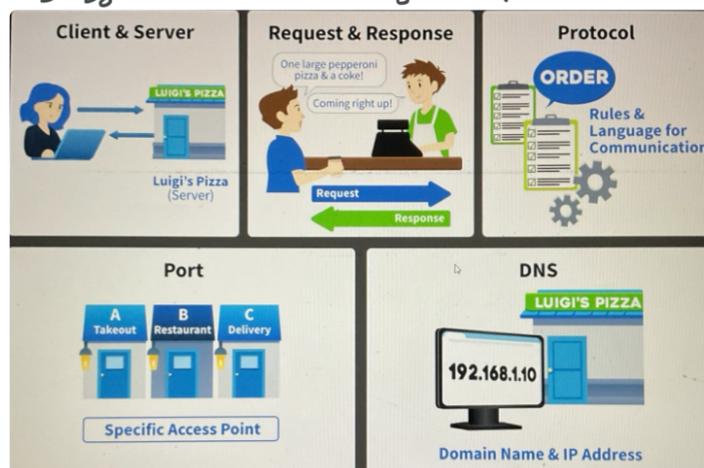
* DNS * Client * Server * Port * Protocol * Network.

Task 2:- Pizza delivery

How computers systems provide services to other systems is to use an analogy: Ex:- how shop offers service to their customers. Let's have a look at pizza takeaway.

It's Friday night, Alice and Bob feel like having pizza. Alice looks at Luigi's pizza's menu and tells Bob which pizza she wants. Bob takes the car, enter Luigi's address into his GPS, and starts driving. Once Bob arrives, he enters Luigi's and places his order: "Get me a large pepperoni pizza and a coke". The employee acknowledges the order and starts making the pizza. Once order is ready, Bob return home and has lovely pizza night with Alice.

This seems like standard, straight forward process. But don't be fooled, because we are so used to ordering pizza, we have internalized the process. Let's take a closer look at each step.



Service, client, server :-

In our analogy, Bob and Alice choose to use pizza takeaway service. Alice is the client, Bob is the server.

In our analogy, Bob is the pizza takeaway service. Alice is client who passes her order to Bob, who then passes it to server at Luigi's Pizzas. He browses to navigate to website. The browser is client that requests webpage, and server is system that serves it.

Request and Response :-

Alice requested a large pizza from Luigi's. Bob didn't make request, was one who bought the request to Luigi's.

Request is not formatted. Generally if requested resource is unavailable, we will get error response.

Luigi's worker responded with order. Ex :- In computer system, we can say Alice used a browser (the client) request to webpage from server, which then sent webpage to client.

Protocol :-

When Alice formulates her request to Bob, she uses language Luigi's pizza understands.

Additionally Alice uses money to determine which orders she can place. Bob understands the

request (language and order) and went Luigi's to bring it, he received a response to understand and brought back to Alice.

A protocol defines how client can communicate with server.

→ Which command do client and server understand Ex :- get command

→ How a request is structured Ex :- First command and then the order.

→ What syntax is used Ex :- Alice uses English language.

→ What response should be given to which type of request Ex :- A request for pizza results in receiving available pizza.

→ What response to give to faulty requests. Ex :- Server at Luigi's pizza says :- No pepperoni pizza available.

Port :-

Port is used to identify specific service running on system. When client want to access a service on server, it must connect using correct port.

Imagine that Luigi's takeaway service requires customers to enter through specific door.

Everyone ordering take away uses same door. In same way, A service on server listens on specific port.

If Luigi's offer multiple services, door A for takeaway, door B for restaurant and door C for delivery.

DNS :-

When Alice sent Bob her request, only name of pizza place was known. With that not possible

to reach destination. So Bob entered Luigi's pizza into GPS device, device returned coordinates for Luigi's.

DNS stand for Domain Name Service and works similarly to GPS. When you enter name for

example:- A website DNS resolves it to server's location. These location coordinates are called

Internet Protocol (IP) addresses in computer term.

Quiz:-

1) What do we use to identify a specific service on server?

A) port

2) What do we call address of a server?

A) Internet protocol address